

Weighted Distance Fitness Function

$$WD = \sum_{i=1}^n d_i \times wc_i$$

d_i = distance of the i -th leg of the route | wc_i = number of remaining deliveries when leg i is traversed

Behavioural insight: a long leg early in the route is penalised heavily because many deliveries still await. Placing short legs first minimises WD.